Converting Units

Name _____ Date _____ No calculator Version 1

KEY WORDS unit · convert · millimetre · kilometre · gram · kilogram · millilitre · litre

A Length ●○○ ANSWER ON THE SHEET

Remember: 10 mm = 1 cm, 100 cm = 1 m, 1000 m = 1 km. Multiply to change to a smaller unit, and divide to change to a larger unit.

1. Fill in the gaps.

a) 4 cm = ___ mm **b)** 3 m = ___ cm **c)** 2 km = ___ m

d) 70 mm = ___ cm **e)** 500 cm = ___ m **f)** 6,000 m = ___ km

g) 2.5 m = ___ cm **h)** 45 mm = ___ cm **i)** 0.8 km = ___ m

j) 320 cm = ___ m **k)** 1,500 m = ___ km **l)** 1.2 m = ___ mm

m) 7.5 cm = ___ mm **n)** 84 cm = ___ m **o)** 250 m = ___ km

p) 0.06 km = ___ m **q)** 3,600 mm = ___ m **r)** 12.5 km = ___ m

B Mass ●○○ ANSWER ON THE SHEET

Remember: 1000 mg = 1 g, 1000 g = 1 kg, 1000 kg = 1 tonne (t).

2. Fill in the gaps.

a) 3 kg = ___ g **b)** 5,000 g = ___ kg **c)** 2 t = ___ kg

d) 4,000 mg = ___ g **e)** 1.5 kg = ___ g **f)** 750 g = ___ kg

g) 0.6 t = ___ kg **h)** 2.4 g = ___ mg **i)** 3,200 kg = ___ t

j) 85 g = ___ kg **k)** 0.25 kg = ___ g **l)** 650 mg = ___ g

C Capacity ●○○ ANSWER ON THE SHEET

Remember: 1000 mL = 1 L, and 1000 L = 1 kL.

3. Fill in the gaps.

a) 2 L = ___ mL **b)** 3,000 mL = ___ L **c)** 5 kL = ___ L

d) 1.5 L = ___ mL **e)** 250 mL = ___ L **f)** 8,000 L = ___ kL

g) 0.75 L = ___ mL **h)** 4,200 mL = ___ L **i)** 0.3 kL = ___ L

j) 60 mL = ___ L **k)** 12.5 L = ___ mL **l)** 450 L = ___ kL

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

D Time ●●○ — ANSWER ON THE SHEET

⚠ **Watch out:** Time does not work in tens: there are 60 seconds in a minute, 60 minutes in an hour and 24 hours in a day.

4. Fill in the gaps.

a) 3 min = ___ s

b) 2 h = ___ min

c) 2 days = ___ h

d) 240 s = ___ min

e) 180 min = ___ h

f) 1.5 h = ___ min

g) 90 min = ___ h

h) 2.5 min = ___ s

i) 72 h = ___ days

j) 0.25 h = ___ min

k) 45 min = ___ h

l) 1.5 days = ___ h

E Problems ●●● — WORK IN YOUR BOOK

Remember: Change every measurement to the same unit before you calculate.

5. Answer each question.

a) A piece of string is 2.4 m long. It is cut into pieces 30 cm long. How many pieces are there?

b) A bottle holds 750 mL. How many litres do 12 bottles hold?

c) A baby has a mass of 3.4 kg. It gains 150 g each week. What is its mass, in kilograms, after 4 weeks?

d) A film lasts 135 minutes. Write this in hours and minutes.

e) Kiri walks 850 m to school and 850 m home again, 5 days a week. How many kilometres does she walk in a week?

f) One tablet holds 500 mg of medicine. How many grams of medicine are in 6 tablets?

★ Extension — FOR EARLY FINISHERS

6. A square with sides of 1 m has sides of 100 cm, so 1 m^2 = 100 × 100 = 10,000 cm^2. How many cm^2 are there in 3 m^2?

7. 1 litre is the same as 1000 cm^3. How many litres does a 5,000 cm^3 container hold?

8. A car travels at 72 km/h. How many metres does it travel in one second?

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

Cell Tuition
1-1 PRIVATE TUTORS

YEAR 9 · MEASUREMENT

Converting Units

Name _____ **Date** _____ No calculator Version 2

KEY WORDS unit · convert · millimetre · kilometre · gram · kilogram · millilitre · litre

A Length ●○○

ANSWER ON THE SHEET

Remember: 10 mm = 1 cm, 100 cm = 1 m, 1000 m = 1 km. Multiply to change to a smaller unit, and divide to change to a larger unit.

1. Fill in the gaps.

a) 6 cm = ___ mm
b) 5 m = ___ cm
c) 4 km = ___ m
d) 90 mm = ___ cm
e) 300 cm = ___ m
f) 8,000 m = ___ km
g) 3.5 m = ___ cm
h) 65 mm = ___ cm
i) 0.4 km = ___ m
j) 480 cm = ___ m
k) 2,500 m = ___ km
l) 1.6 m = ___ mm
m) 9.5 cm = ___ mm
n) 72 cm = ___ m
o) 750 m = ___ km
p) 0.09 km = ___ m
q) 4,800 mm = ___ m
r) 10.5 km = ___ m

B Mass ●○○

ANSWER ON THE SHEET

Remember: 1000 mg = 1 g, 1000 g = 1 kg, 1000 kg = 1 tonne (t).

2. Fill in the gaps.

a) 4 kg = ___ g
b) 7,000 g = ___ kg
c) 3 t = ___ kg
d) 6,000 mg = ___ g
e) 2.5 kg = ___ g
f) 250 g = ___ kg
g) 0.8 t = ___ kg
h) 1.8 g = ___ mg
i) 4,600 kg = ___ t
j) 65 g = ___ kg
k) 0.75 kg = ___ g
l) 450 mg = ___ g

C Capacity ●○○

ANSWER ON THE SHEET

Remember: 1000 mL = 1 L, and 1000 L = 1 kL.

3. Fill in the gaps.

a) 4 L = ___ mL
b) 6,000 mL = ___ L
c) 2 kL = ___ L
d) 2.5 L = ___ mL
e) 750 mL = ___ L
f) 9,000 L = ___ kL
g) 0.25 L = ___ mL
h) 3,600 mL = ___ L
i) 0.7 kL = ___ L
j) 80 mL = ___ L
k) 10.5 L = ___ mL
l) 350 L = ___ kL

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

D Time ●●○ — ANSWER ON THE SHEET

⚠ **Watch out:** Time does not work in tens: there are 60 seconds in a minute, 60 minutes in an hour and 24 hours in a day.

4. Fill in the gaps.

a) 5 min = ___ s

b) 4 h = ___ min

c) 3 days = ___ h

d) 360 s = ___ min

e) 300 min = ___ h

f) 2.5 h = ___ min

g) 150 min = ___ h

h) 1.5 min = ___ s

i) 96 h = ___ days

j) 0.75 h = ___ min

k) 15 min = ___ h

l) 2.5 days = ___ h

E Problems ●●● — WORK IN YOUR BOOK

💡 **Remember:** Change every measurement to the same unit before you calculate.

5. Answer each question.

a) A piece of string is 3.6 m long. It is cut into pieces 40 cm long. How many pieces are there?

b) A bottle holds 600 mL. How many litres do 15 bottles hold?

c) A baby has a mass of 3.2 kg. It gains 200 g each week. What is its mass, in kilograms, after 4 weeks?

d) A film lasts 165 minutes. Write this in hours and minutes.

e) Kiri walks 650 m to school and 650 m home again, 5 days a week. How many kilometres does she walk in a week?

f) One tablet holds 250 mg of medicine. How many grams of medicine are in 8 tablets?

★ Extension — FOR EARLY FINISHERS

6. A square with sides of 1 m has sides of 100 cm, so 1 m^2 = 100 × 100 = 10,000 cm^2. How many cm^2 are there in 5 m^2?

7. 1 litre is the same as 1000 cm^3. How many litres does a 8,000 cm^3 container hold?

8. A car travels at 54 km/h. How many metres does it travel in one second?

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.